Pierre Fernandez

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Research Scientist specializing in watermarking, provenance and responsible generative AI. I develop methods to detect, attribute, and trace AI-generated content across modalities (text, image, video, audio).

Experience

Meta, *Research Scientist* **2024 – present**
FAIR (Fundamental AI Research), Watermarking team *Paris, France*
Primary contributor to watermarking research and production efforts.

- **Research**: co-led TextSeal, a state-of-the-art LLM watermark with stronger detection than SynthID-text and localization in mixed human/AI documents; built state-of-the-art image and video watermarking models (Pixel Seal, Video Seal)
- **Exploratory research**: watermarking autoregressive image generation; geometric image synchronization with deep watermarking; post-hoc text watermarking through LLM rephrasing; asymmetric (public-key) watermark detection (ongoing)
- **Open source**: released Content Seal, including TextSeal, VideoSeal (500+ stars), and Watermark-Anything (1000+ stars)
- **Internal impact**: automated copy detection in the IG feed ($1.4M/yr savings); models deployed in Muse Image (MSL, public detector at meta.ai/identification), FB videos, MovieGen, and Imagine for compliance and detection of AI-generated content
- **Policy**: member of the expert group for the EU AI Act code of practice (reviewed drafts); technical advisor to Meta's policy teams
- **Team building**: interviewed and recruited candidates, growing the watermarking team from 0 to 8 people in one year
- **Mentoring**: 1 postdoc, 1 PhD student (both now in the team); supervised 2 research interns (1 NeurIPS paper)

Meta & Inria, *PhD student* **2021 – 2024**
FAIR (Fundamental AI Research) & Linkmedia team at Inria *Rennes & Paris, France*
Started the watermarking research at Meta as part of the FAIR Paris PhD program.

- **Research**: first author of Stable Signature, AudioSeal (500+ stars), statistical tests for LLM watermark detection (best student paper award), and radioactive watermarking; trained and fine-tuned 1B+ parameter models across modalities
- **Internal impact**: watermarking for Meta Imagine, Seamless, and Audiobox (Instagram auto-dubbing)
- **Other**: developed video and image classification fine-tuning and evaluations for DINOv2

Dassault Systèmes, *Research Intern* **Apr. 2020 – Aug. 2020 & Sep. 2021 – Dec. 2021**
Differentiable physics (Jax, Taichi) for robotics and structural optimization *Vélizy, France*

Incepto Medical, *Software Engineer Intern* **Jul. 2019 – Sep. 2019**
Visualization tool for medical-imaging deep learning results (AngularJS) *Station F - Paris, France*

Education

Meta & Inria, *PhD in Mathematics and Computer Science* **2021 – 2025**
Watermarking across Modalities for Content Tracing and Generative AI (defended Jan. 2025) *Rennes & Paris, France*
Advisors: Teddy Furon (Inria), Hervé Jégou (FAIR), Matthijs Douze (FAIR)

Paris-Saclay University, *MSc in Mathematics, Vision and Learning ("MVA")* **2020 – 2021**
Highest Honors, GPA: 4.0/4.0 *Paris, France*

École polytechnique, *BSc & MScT in Artificial Intelligence and Visual Computing* **2017 – 2021**
Highest Honors, GPA: 3.95/4.0 *Palaiseau, France*
Awards: Research Internship Award (top 5%), Outstanding Investment and Leadership awards; ranked 4th at entrance exams

Invited Talks

CVPR APAI Workshop (2026), **X-IA** (2025), **BLISS AI, TU Berlin** (2025), **AAAI Defactify Workshop** (2025), **Geneva Institute**, **Sony CSL**, **Photoroom**, **Inria**, **Qualcomm AI** and others

Additional Information

Languages	French (native), English (working proficiency: TOEFL 110/120, TOEIC 970/990), Spanish (working proficiency)
Programming	Python (PyTorch, distributed training with FSDP, Jax), Slurm; basic JavaScript, SQL, C++, Java
Community	Reviewer for CVPR, ICCV, ICLR, ICML, IEEE TIFS, IJCV, JMLR, Multimedia Forensics Co-organized the GenAI Watermarking workshop (ICLR 2025), a watermarking tutorial (CVPR 2026), and a special session on watermarking generative models (ACM IH&MMSec 2026)
Misc.	Jury for GenAI start-up pitch (VivaTech 2023), advisor for Kartoon.ai (Station F acceleration program)